```
from google.colab import files
uploaded = files.upload()
```

Choose Files Screenshot… 133747.png

Screenshot 2026-08-07 133747.png(image/png) - 142757 bytes, last modified: 8/7/2026 - 100% done

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Saving Screenshot 2026-08-07 133747.png to Screenshot 2026-08-07 133747.png
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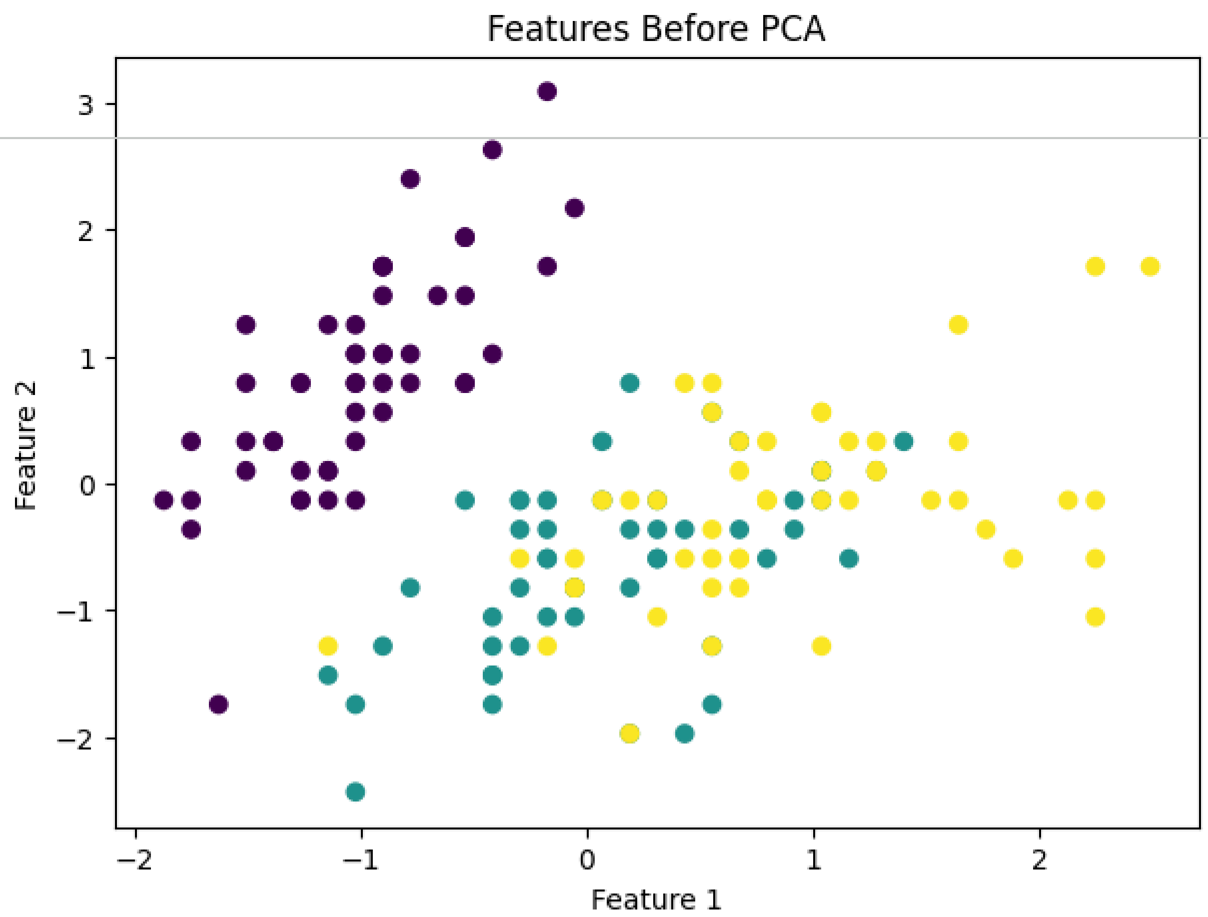
```
# PCA Visualization Study

import numpy as np
import matplotlib.pyplot as plt
from sklearn.datasets import load_iris
from sklearn.preprocessing import StandardScaler
from sklearn.decomposition import PCA

# 1. Load dataset
iris = load_iris()
X = iris.data
y = iris.target

# 2. Standardize the features
scaler = StandardScaler()
X_scaled = scaler.fit_transform(X)

# 3. Visualization BEFORE PCA
plt.figure(figsize=(7, 5))
plt.scatter(X_scaled[:, 0], X_scaled[:, 1], c=y)
plt.xlabel("Feature 1")
plt.ylabel("Feature 2")
plt.title("Features Before PCA")
plt.show()

# 4. Apply PCA
pca = PCA(n_components=2)
X_pca = pca.fit_transform(X_scaled)
```

```
# 5. Explained variance
print("Explained Variance Ratio:")
print(pca.explained_variance_ratio_)

print("\nTotal Variance Preserved:",
      sum(pca.explained_variance_ratio_))

# 6. Visualization AFTER PCA
plt.figure(figsize=(7, 5))
plt.scatter(X_pca[:, 0], X_pca[:, 1], c=y)
plt.xlabel("Principal Component 1")
plt.ylabel("Principal Component 2")
plt.title("Features After PCA")
plt.show()

# 7. Display transformed data
print("\nFirst 5 PCA-transformed samples:")
print(X_pca[:5])
```




```
Explained Variance Ratio:
```